

MD KAMRUZZAMAN

Python Backend Developer | FastAPI | PostgreSQL | Docker
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Python backend developer with a B.Tech in Computer Science and Engineering with a Machine Learning specialization. Hands-on experience building production-style REST APIs using FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT authentication, pytest, Docker, and GitHub Actions. Experienced in database design, API development, automated testing, containerization, and CI workflows. Currently expanding into MLOps through model training pipelines, MLflow experiment tracking, FastAPI model serving, and Docker.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java (working knowledge)
- **Backend:** FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic
- **Databases:** PostgreSQL, MySQL
- **Authentication & Security:** JWT, bcrypt
- **Testing & DevOps:** pytest, Docker, GitHub Actions, CI, Git, Linux
- **Data & ML:** NumPy, Pandas, Scikit-learn, Machine Learning
- **MLOps:** MLflow, Model Training Pipelines, Model Serving, Docker

PROJECTS

Expense Tracker API

[GitHub](#)

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT, Docker, pytest, GitHub Actions

- Built and deployed a REST API for personal expense tracking with JWT-based authentication and bcrypt password hashing for protected endpoints.
- Implemented user-level data isolation so authenticated users can access only their own expense records at the database query layer.
- Designed a PostgreSQL data model using SQLAlchemy ORM and managed database schema changes with Alembic migrations.
- Implemented pagination, category and amount filtering, request validation, and structured HTTP error handling using FastAPI and Pydantic.
- Wrote automated pytest tests covering authentication, CRUD operations, authorization, validation, and edge cases.
- Containerized the application with Docker and configured GitHub Actions to automatically run the test suite and code-quality checks.
- Deployed the backend using Render with PostgreSQL hosted on Neon, making the API publicly testable.

ML Expense Platform

[GitHub](#)

Python, Scikit-learn, MLflow, FastAPI, Docker

- Built an end-to-end machine learning workflow covering model training, experiment tracking, and API-based model serving.
- Integrated MLflow to track machine learning experiments and model development workflows.
- Implemented a FastAPI `/predict` endpoint for serving trained machine learning models through a REST API.
- Containerized the ML application with Docker to provide a reproducible runtime environment.
- Structured the project to demonstrate practical integration of machine learning, backend engineering, and MLOps concepts.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Completed 2025

Lovely Professional University, Punjab, India
Specialization: Machine Learning

CERTIFICATIONS & RELEVANT LEARNING

- **Meta Back-End Developer** (Coursera) – [Verify](#)
- **Python Data Structures** (University of Michigan, Coursera) – [Verify](#)
- **Java Programming Fundamentals** (Coursera) – [Verify](#)
- **Natural Language Processing Specialization** (DeepLearning.AI, Coursera) – [Verify](#)